

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

5. Structured Knowledge Representation – Solved Problems

Every problem opens with the question **exactly as the paper prints it**, the paper's own spelling and slips included. · 11 sentence sets over 12 papers, plus the net-to-frame conversion and the net/frame/FOPL question. · Where a lecture deck or the textbook already draws the answer, the published figure is printed and the caption says exactly what differs from the paper's wording. Where none does, the answer is given as an arc list — every node, every link, every direction — from which the drawing is mechanical.

The four-step recipe. It works on all eleven sets:

- **Step 1. Underline every noun phrase.** Each distinct one is a **node**. Do not invent nodes and do not merge two that the paper keeps apart.
- **Step 2. Sort the nodes into classes and individuals.** A proper name, or anything the sentences call *a* something, is an **individual**. Anything that other things can *be* is a **class**.
- **Step 3. Draw the class spine first, bottom to top**, joined by *isa* arcs, then hang each individual off its class with an *instance* arc.
 - **Class to class is isa. Individual to class is instance.** Getting these two the wrong way round is the single commonest lost mark in this question.
- **Step 4. Add one labelled arc per remaining sentence**, always *out of* the thing the sentence is about. The verb becomes the arc label.
- **Then check three things**, which is what separates a 6 from a 4:
 - **Count.** One arc per sentence, plus the spine. If a sentence produced no arc you have lost a mark.
 - **Push properties up.** If two individuals of the same class both carry a property, draw it **once on the class** and let them inherit. Say in one line that you did, and why.
 - **Flag what the net cannot say.** Quantifiers, negation and disjunction do not draw. A sentence like “*every mammal gives birth to a baby*” gets an arc out of the class node and **a note saying the arc is read as $\forall x \in Mammal$** . Examiners give marks for that note.
- **Layout that always works:** most general class at the **top**, *isa* arcs pointing **up**, individuals on a row near the bottom, and their own properties fanning **sideways**. Every published figure in this chapter is drawn that way.

5.N.1 A person is a mammal: Sakti Gauchan, and the same set as Sandeep Lamichhane

Convert the given sentences into a semantic net 7 2+6

PIN 2/41 72 Ma 7, 78 Po 2+6 · two papers, **one net**: 78 Po is 72 Ma with the cricketer's name changed · both also ask “*what are frames and semantic net*” first, which is 5.6 of the theory

QUESTION AS PRINTED 72 Ma · Q6 · 7

What are Frames and Semantic Net? Convert the given sentences in Semantic Net.

- i) A person is a mammal.
- ii) Sakti Gauchan is a person.
- iii) Person has nose.
- iv) Sakti Gauchan is in Nepalese team.
- v) Uniform color of sakti Gauchan is Red/Blue.

QUESTION AS PRINTED 78 Po · Q6 · 2+6

What are Frames and Semantic Net? Convert the given sentences in Semantic Net.

- A person is a mammal.
- Sandeep Lamichhane is a person.
- Person has nose.
- Sandeep Lamichhane is in Nepalese cricket team.
- **Uniform color of Sakti Gauchan is Red/Blue.**

The fifth sentence names a *different* cricketer: the paper was copied from the 72 Ma set and this one line was never updated. Reproduced as printed.

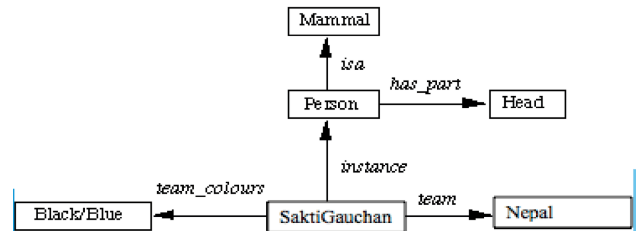
Step 1–2: nodes

- **Classes:** Mammal, Person, Nepalese team (a named team, so it can also be drawn as an individual – either is accepted).
- **Individual:** Sakti Gauchan.
- **Values:** Nose, Red/Blue.

Step 3–4: the arcs

From	Link	To
Person	isa	Mammal
Sakti Gauchan	instance	Person
Person	has-part	Nose
Sakti Gauchan	team	Nepalese team
Sakti Gauchan	uniform-color	Red/Blue

- **5 sentences, 5 arcs.** The count checks.
- **Note to write beside the drawing:** *has-part* sits on **Person**, not on Sakti Gauchan, so **every** person inherits a nose. That is the inheritance the question is testing.
- 78 Po: identical net, *Sakti Gauchan* replaced by *Sandeep Lamichane* in the first four arcs. **For the fifth, draw the uniform-colour arc off Sandeep and add one line:** “the paper names Sakti Gauchan here; taken as the same person the other four sentences are about.” Say it and you lose nothing.



The published answer to this exact question, from PS Sir's own slide. Two differences from the paper's wording, both harmless: it draws *has-part* → *Head* where the paper says *nose*, and *team_colours* → *Black/Blue* where the paper says *Red/Blue*. Substitute the paper's words; the shape is unchanged. Source: PS Sir ch5.

The FOPL the deck writes underneath

- `ISA(Mammal, Person),`
`INSTANCE(Person, Sakti Gauchan),`
`HAS_PART(Person, head),`
`TEAM(Nepal, Sakti Gauchan),`
`TEAM_COLOURS(black/blue, Sakti Gauchan).`
- **The deck is inconsistent, twice over.** An earlier slide writes `MAMMAL(person)` for the first sentence – a *unary* predicate that loses the relation entirely. And it writes the arguments `(class, individual)` here, the reverse of Rich and Knight's `instance(Pee-Wee-Reese, Person)`.
- **Use one order and say which.** This file writes `isa(subclass, superclass)` and `instance(individual, class)`, matching the arrow direction in every drawing. **Consistency is what is marked, not the choice:** pick one, state it in a line, and never swap.

5.N.2 The baseball player: Pee-wee-Reese, and the same set as Rohit Paudel

Convert the given sentences into a semantic net 7 2+6

PIN 2/41 73 Bh 7, 79 Ch 2+6 · again **two papers, one net:** 79 Ch is 73 Bh moved from baseball to cricket · this is Rich and Knight's own figure, so **it is also the set to expect in a “convert the net into a frame” question**

QUESTION AS PRINTED 73 Bh · Q6 · 7

Convert given sentences into Semantic Network.

- i) The height of the adult male is 5.10
- ii) Baseball player is an adult male.
- iii) Adult male is a person.
- iv) Batting average of Baseball players is 0.252
- v) Pee-wee-Reese is a Fielder.
- vi) Fielder is Baseball player.
- vii) Team of pee-wee-Reese is Brooklyn Dodger.

QUESTION AS PRINTED 79 Ch · Q4 · 2+6

What is semantic network? Convert the following sentences into semantic network.

- i) The avg height of an adult is 5'6".
- ii) Cricket player is an adult male.
- iii) Adult male is a person.
- iv) Batting average of a cricket player is 37.59

- v) Rohit Paudel is a Fielder.
- vi) Fielder is a cricket player.
- vii) Team of Rohit Paudel is Nepal.

Sentence (i) says *adult* where (ii) and (iii) say *adult male*. Read as the same node.

The spine is four deep, and that is the whole difficulty

From	Link	To
Adult Male	isa	Person
Baseball Player	isa	Adult Male
Fielder	isa	Baseball Player
Pee-wee-Reese	instance	Fielder
Adult Male	height	5.10
Baseball Player	batting-average	0.252
Pee-wee-Reese	team	Brooklyn Dodger

- **7 sentences, 7 arcs.** The count checks.
- **The mark is in where the properties hang.** *height* goes on **Adult Male** and *batting-average* on **Baseball Player**, because that is the class each sentence names. Hanging either off Pee-wee-Reese loses the inheritance the question is set to test.
- **Read the inheritance back** in one line: Pee-wee-Reese is a Fielder, so a Baseball Player, so an Adult Male, so a Person – and therefore **inherits height 5.10 and batting average 0.252 without either being stored on him.**
- 79 Ch: same seven arcs. Baseball Player → Cricket Player, 5.10 → 5'6", 0.252 → 37.59, Pee-wee-Reese → Rohit Paudel, Brooklyn Dodger → Nepal.

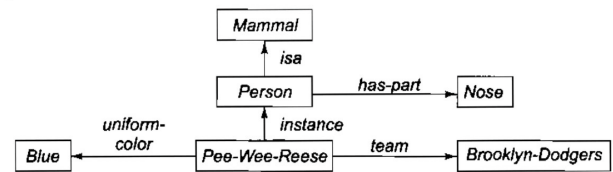


Fig. A Semantic Network

isa(Person, Mammal)
instance(Pee-Wee-Reese, Person)
team (Pee-Wee-Reese, Brooklyn-Dodgers)
uniform-color(Pee-Wee-Reese, Blue)

The figure both papers are drawn from, in its shorter textbook form: it keeps *Person*, *Pee-Wee-Reese* and the *team* arc but collapses the *Adult Male* / *Baseball Player* / *Fielder* spine, and its property arcs are *has-part* and *uniform-color*. **Copy the shape, use the seven arcs in the table** for the exam answer. Source: PS Sir ch5.

Why this set is worth extra attention

- It is the only sentence set in the papers with **four levels of class**, so it is the one that shows inheritance properly.
- It is also **the set** 81 Ba **converts into a frame system** – see §5.N.10, where the finished frames are printed.

5.N.3 Tom the cat: two sentence sets, three papers

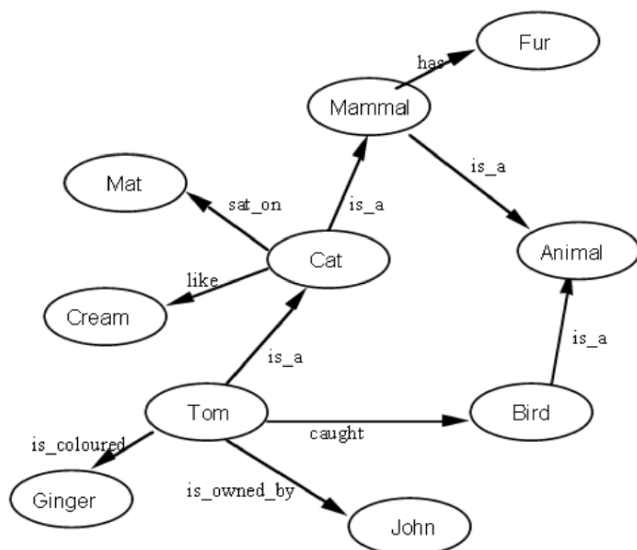
Represent the following sentences using semantic net 2+5 2+6

PIN 3/41 82 Bh 2+5, 81 Ch 2+6, 79 Bh 2+6 · **the most-repeated set in the chapter** · 82 Bh and 81 Ch print it word for word identically; 79 Bh is a rewritten variant · all three pair it with a KR short answer, *issues in KR* twice and *define semantic net* once

QUESTION AS PRINTED 82 Bh Q5 2+5 · 81 Ch Q5 2+6 · the same eleven sentences

What are different issues in knowledge representation? Represent the following sentences using semantic net representation.

Tom is a cat. Tom caught a bird. Tom is owned by Sudeep. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammals have fur. Sudeep age is 20.



The published answer, ten of the eleven arcs already drawn. It is the textbook set, so the only difference from the paper is the owner's name: *John* in the figure, *Sudeep* in the paper. The eleventh sentence, *Sudeep age is 20*, is not in the figure – add one arc, *Sudeep age* → 20. Source: BA Sir CH-05.

The eleven arcs, in the order the sentences give them

From	Link	To
Tom	instance	Cat
Tom	caught	Bird
Tom	is-owned-by	Sudeep
Tom	is-coloured	Ginger
Cat	like	Cream
Cat	sat-on	Mat
Cat	isa	Mammal
Bird	isa	Animal
Mammal	isa	Animal
Mammal	has	Fur
Sudeep	age	20

- **11 sentences, 11 arcs.** The count checks.
- **Two traps, and both are worth a mark each:**
 - *Tom is a cat* is instance (Tom is an individual); *a cat is a mammal* is isa (both are classes). Same English construction, different link.
 - *The cat sat on the mat* is about **the class Cat** as the figure draws it, but the sentence plainly means *that cat*, ie Tom. **Draw it as the figure does and add a line saying the natural reading is Tom** – the ambiguity is exactly the “lack of formality” limitation the paired theory question asks about.
- *Cats like cream* and *mammals have fur* are drawn out of the class nodes, so Tom inherits both. Say so.

The 2079 Bhadra variant: eleven different sentences, same shape 2+6

79 Bh 2+6 · counted in the chip above · **the last sentence is the interesting one:** it is a quantified statement, which a semantic net cannot properly express

QUESTION AS PRINTED 79 Bh · Q6 · 2+6

List the issues need to be consider in Knowledge Representation techniques. Convert given sentences into Semantic Network.

- a) Tom is a cat b) Tom fights with rat c) Tom is owned by Ram
- d) Tom is black in color. e) Cats like milk f) Rats like cheese
- g) The cat sat on the bed h) A cat is a mammal i) A Rat is an animal
- j) All mammals are animals k) Every mammal gives birth a baby

From	Link	To
Tom	instance	Cat
Tom	fights-with	Rat
Tom	is-owned-by	Ram
Tom	is-coloured	Black
Cat	like	Milk
Rat	like	Cheese
Cat	sat-on	Bed
Cat	isa	Mammal
Rat	isa	Animal
Mammal	isa	Animal
Mammal	gives-birth-to	Baby

- **11 sentences, 11 arcs.** The count checks.
- Layout: *Animal* at the top, *Mammal* and *Rat* under it, *Cat* under *Mammal*, *Tom* at the bottom, and every value node (Milk, Cheese, Bed, Black, Ram, Baby) hanging sideways.
- **(k) is the one that earns the extra mark.** “Every mammal gives birth to a baby” is $\forall x (x \in \text{Mammal} \Rightarrow \exists y (y \in \text{Baby} \wedge \text{gives-birth-to}(x, y)))$.
 - The net can only draw **one arc** out of the *Mammal* class node.
 - **Write underneath:** “the arc out of a class node is read as ‘for every member’, and the existential ‘some baby’ is not expressible in a semantic net at all – this is the quantification limitation.”
- *Tom fights with rat* is between Tom and the class Rat as printed. Note that a fight is with one particular rat, so a strict reading needs an **event node** (§5.4 of the theory).

5.N.4 Mammals, fur and water

Define term frame. Convert the following sentences into a semantic network 1+6

PIN 1/41 82 Ka · 1+6 · the set with three siblings under one class, so it is the one that rewards pushing a property up · the 1-mark opener is *define a frame*, \$5.5 of the theory, one sentence

QUESTION AS PRINTED 82 Ka · Q5 · 1+6

Define term frame. Convert the following sentences into semantic network. Cat is a mammal. Cat has fur. Mammal has vertebra. Mammal is an animal. Bear is a mammal. Bear has fur. Whale is a mammal. Whale lives in water. **Fish is an animal 10.** Fish lives in water.

The “10.” is the paper's own stray list number printed inside the sentence. Reproduced as printed; it is not a tenth sentence.

From	Link	To
Cat	isa	Mammal
Bear	isa	Mammal
Whale	isa	Mammal
Mammal	isa	Animal
Fish	isa	Animal
Cat	has	Fur
Bear	has	Fur
Mammal	has	Vertebra
Whale	lives-in	Water
Fish	lives-in	Water

- **10 sentences, 10 arcs.** The count checks.
- **Every node here is a class**, so **every** link is *isa* or a property arc. There is not one *instance* arc in this set — no individual is named. Saying that in one line shows you understood the difference rather than guessed it.

Layout

- *Animal* on the top row. *Mammal* and *Fish* on the second. *Cat*, *Bear*, *Whale* on the third. Values (*Fur*, *Vertebra*, *Water*) off to the right.
- *Water* is one node reached by two arcs, not two nodes. **A semantic net stores each object once**; drawing it twice is the mistake that costs a mark here.

The observation that earns the last mark

- *Cat has fur* and *Bear has fur* are both given, and both are drawn because the question says to convert the sentences.
- But fur is a **mammal** property. Add a line: “*has fur* could be moved onto *Mammal*, and then *Cat*, *Bear* and *Whale* would all inherit it — that is the space saving inheritance buys.”
- **Then give the counter-example in the same set**: whales are mammals and do *not* have fur in the ordinary sense, so moving it up would make the net assert something false. **A net has no way to state an exception**, which is precisely why frames carry defaults and overrides.
- That pair of sentences is the best two-line answer to “*limitations of semantic nets*” anywhere in these papers, and it comes free with this question.

5.N.5 Fido the dog

Explain the approaches of KR. Highlight pros and cons of semantic nets and frames. Use them to represent the following sentences 2+6

PIN 1/41 81 Bh · 2+6 · the smallest set in the papers, six sentences · it asks for both representations, so draw the net *and* write the frames — the arc list below does both

QUESTION AS PRINTED 81 Bh · Q6 · 2+6

Explain different approaches of knowledge representation. Highlight pros and cons of Semantic Nets and Frames. Use them to represent following sentences.

- **Figo is a dog.**
- Fido is own by Rohit.
- Fido is ginger in color.
- Dog is a mammal.
- All Mammals are animals.
- Mammals have fur.

The paper names the dog *Figo* in the first line and *Fido* in the next two. Reproduced as printed; take them as one animal and say so.

As a semantic net

From	Link	To
Fido	instance	Dog
Fido	is-owned-by	Rohit
Fido	is-coloured	Ginger
Dog	isa	Mammal
Mammal	isa	Animal
Mammal	has	Fur

- **6 sentences, 6 arcs.** The count checks.
- Spine: *Animal* ← *Mammal* ← *Dog*, with *Fido* hung off *Dog*. *Rohit* and *Ginger* sideways.
- **Open the answer with one line on the name:** “the paper writes *Fido* once and *Fido* twice; taken as the same dog throughout.” It costs nothing and shows you read the question.
- *Mammals have fur* is on the class, so **Fido inherits fur** without it being stored on him. Write that sentence out – this set is small enough that the inheritance remark is where the marks are.

The same knowledge as frames

Frame	Slot	Filler
Animal	(class frame)	–
Mammal	isa	Animal
	has	Fur
Dog	isa	Mammal
Fido	instance	Dog
	owner	Rohit
	colour	Ginger
	has	Fur (inherited)

- **One frame per node, one slot per outgoing arc**, exactly the recipe in §5.6 of the theory.
- **Write the inherited row in italics or brackets** and label it *inherited*. It shows the frame system doing the work, and it is the difference between copying the net out as a table and answering the question.
- **Pros and cons** (the first half of the question) are the two lists in §5.4 and §5.5 of the theory. One line each is enough at 2 marks.

5.N.6 Employees and supervisors

When do we use a semantic network? Convert the following sentences into a semantic network 2+6

PIN 1/41 82 Ba 2+6 · **the trickiest set in the papers:** it contains a word that is both a class and a relation, and a symmetric relation · the 2-mark opener, *when do we use a semantic net*, is §5.4 of the theory

QUESTION AS PRINTED 82 Ba · Q6 · 2+6

When do we use semantic network? Convert the following sentences into semantic network.

- i) Employee is a human. ii) Hari is an employee. iii) Sita is a human.
- iv) Supervisor is a human. v) Ram is a supervisor. vi) Ram is married to Sita.
- vii) Ram is supervisor of Hari.

From	Link	To
Employee	isa	Human
Supervisor	isa	Human
Hari	instance	Employee
Sita	instance	Human
Ram	instance	Supervisor
Ram	married-to	Sita
Ram	supervisor-of	Hari

- **7 sentences, 7 arcs.** The count checks.
- Layout: *Human* at the top; *Employee* and *Supervisor* under it; *Hari*, *Ram*, *Sita* on the bottom row so the two arcs between individuals are horizontal and easy to read.

Three things the examiner is testing here

- **1. Supervisor is used two ways in one question.**
 - In (iv) and (v) it is a **class node**, reached by *isa* and *instance*.
 - In (vii) *supervisor-of* is a **relation arc between two individuals**, Ram to Hari. **It is not the class node.**
 - Drawing an arc from Hari to the *Supervisor* class is the mistake this set is built to catch.
- **2. Sita is an instance of Human directly**, not of *Employee* or *Supervisor*. The sentences never say what she does, so do not invent a class for her. **Adding information the sentences do not give loses marks**, however sensible it looks.
- **3. married-to is symmetric.** A semantic net arc is directed, so either draw a **double-headed** arrow or add the inverse arc *Sita married-to* → *Ram*, and say which convention you used.
 - This is the “*relationships among attributes*” issue from §5.3: inverse and symmetric relations have to be handled explicitly, because the formalism does not know they exist.

5.N.7 FOPL first, then the net

Convert the following sentences into FOPL and represent using a semantic network 2+6

PIN 1/41 81 Ash 2+6 · the only set that asks for FOPL as well as the drawing, so it is worth doing every set this way in practice · the FOPL is free marks: it is the arc list written as predicates

QUESTION AS PRINTED 81 Ash · Q6 · 2+6

What are Frames and Semantic Network? Convert the following sentences into FOPL and represent using semantic network.

- i) Ram is a person. ii) Person is an animal. iii) Ram likes Jhilke. iv) Jhilke is a cat.
- v) Cat eats fish. vi) Cat is an animal. vii) fish is an animal.

Arc list and FOPL, side by side

From	Link	To	FOPL
Ram	instance	Person	instance(Ram, Person)
Person	isa	Animal	isa(Person, Animal)
Ram	likes	Jhilke	likes(Ram, Jhilke)
Jhilke	instance	Cat	instance(Jhilke, Cat)
Cat	eats	Fish	see below
Cat	isa	Animal	isa(Cat, Animal)
Fish	isa	Animal	isa(Fish, Animal)

- **7 sentences, 7 arcs.** The count checks.
- **The rule for turning an arc into FOPL:** an arc $A \xrightarrow{R} B$ becomes the binary predicate $R(A, B)$, with **isa** used between classes and **instance** between an individual and its class. Nothing else is needed for six of the seven.

Sentence (v) is the one worth thinking about

- “Cat eats fish” is a statement about **every** cat and **some** fish, so its honest FOPL is

$$\forall x (Cat(x) \Rightarrow \exists y (Fish(y) \wedge eats(x, y)))$$

- The net can only draw **one arc**, $Cat \text{ eats } \rightarrow Fish$, class to class.
- **Write both, and say what is lost:** the arc records the relation but throws away the quantifiers. That single remark is the entire “a semantic net cannot express quantification” limitation, evidenced on the examiner’s own data.
- If you want the flat version for the arc table, `eats(Cat, Fish)` is what every deck writes and is accepted.

Layout

- *Animal* on top, with **three** `isa` arcs into it, from *Person*, *Cat* and *Fish*.
- *Ram* under *Person*, *Jhilke* under *Cat*, and the `likes` arc running horizontally between the two individuals.
- **The net has a nice shape here:** two parallel individual-to-class chains joined at the top by *Animal* and at the bottom by `likes`. Draw it that way and the seven arcs never cross.

5.N.8 Birds: Tweety and Sweety

What are frames and semantic nets? Convert the given sentences into a semantic net 2+6

PIN 1/41 77 Ch 2+6 · two sentences here need more than one arc each, which is what makes this set harder than it looks · the same paper asks conceptual dependency and scripts at Q 8, so §5.7 of the theory is on the same sheet

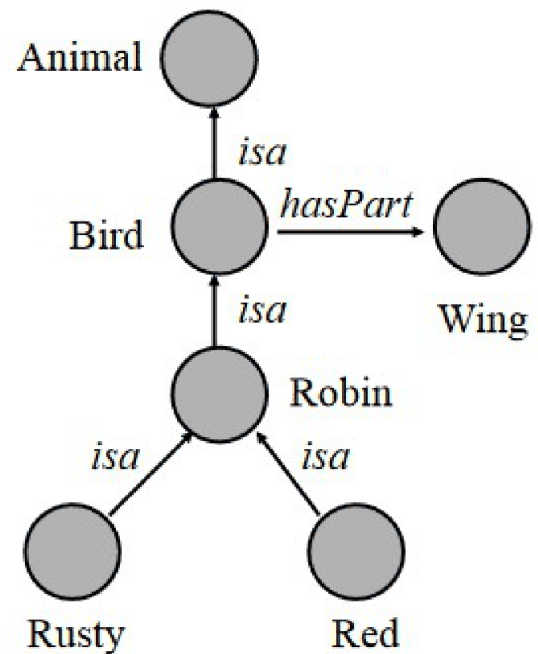
QUESTION AS PRINTED 77 Ch · Q6 · 2+6

What are Frames and Semantic Net? Convert the given sentences in semantic Net:

- Tweety and Sweety are birds. Tweety has a red beak. Sweety is Tweety’s child.
- A crow is a bird. Birds can fly. Sparrow is a bird. Sparrow has a wing.

From	Link	To
Tweety	instance	Bird
Sweety	instance	Bird
Tweety	has-part	Beak
Beak	is-coloured	Red
Sweety	child-of	Tweety
Tweety	parent-of	Sweety (inverse)
Crow	isa	Bird
Sparrow	isa	Bird
Bird	can	Fly
Sparrow	has-part	Wing

- **7 sentences, 10 arcs.** Here the count does *not* match one to one, and that is the point:
 - *Tweety and Sweety are birds* is **two** `instance` arcs, one per bird.
 - *Tweety has a red beak* is **two** arcs: the beak is a part of Tweety, and *red* is a property of the **beak**, not of Tweety. **Drawing one arc *Tweety* red-beak → Red loses the mark** – the colour belongs to the part.
 - *Sweety is Tweety's child* is one arc, but the **inverse** arc is worth drawing and labelling.



The shape to copy, from PS Sir's deck: *Animal* ← *Bird* ← *Robin* ← *Rusty* / *Red*, with `hasPart` → *Wing* drawn once on the *Bird* class. Substitute this paper's nodes. Source: PS Sir ch5.

- **Note to add:** *Sparrow has a wing* is drawn as the paper gives it, on *Sparrow*, but wings are a property of **all** birds – moving `has-part` → *Wing* up to the *Bird* class would let Tweety, Sweety, Crow and Sparrow inherit it from one arc.
- *Birds can fly* is on the class, so every bird inherits it. That is also the standard **exception** example: a penguin is a bird and cannot fly, and a semantic net has no way to override an inherited property. Frames do, with a default.

5.N.9 A semantic net about the AI course itself

Draw a sample net about your course AI to demonstrate the features and qualities of a semantic net **7**

PIN 1/41 72 Ma **7** · **the only open-ended drawing question in the papers:** the content is yours, the marks are for demonstrating the *features* · so build the net so that each feature appears at least once, and label them

QUESTION AS PRINTED 72 Ma · Q5 · **7**

What is a semantic net? Draw a sample net about your course AI to demonstrate the knowledge representation features and qualities of semantic net.

A net that hits every feature

From	Link	To
Elective	isa	Course
AI (CT 710)	instance	Course
Course	has-part	Chapter
AI (CT 710)	credit-hours	3
AI (CT 710)	semester	IV/I
AI (CT 710)	programme	BEI
Search	part-of	AI (CT 710)
A*	isa	Search
Student	studies	AI (CT 710)
Ram	instance	Student
AI (CT 710)	taught-by	Teacher
Teacher	isa	Person
Student	isa	Person

Label the features on the drawing, one arrow each

- **Class hierarchy** – *Person* ← *Student* / *Teacher*, joined by *isa*.
- **Instances** – *Ram* *instance* → *Student*. Point out that *isa* and *instance* are different links.
- **Inheritance** – anything put on *Person* is available to *Ram* through two arcs. Write the chain out.
- **Part-whole** – *Search* *part-of* → *AI*, which does **not** inherit the way *isa* does. Saying that is worth a mark on its own.
- **Attribute arcs** – *credit-hours*, *semester*, *programme*: values rather than objects.
- **Intersection search** – start from *Ram* and from *A** and the two wavefronts meet at *AI (CT 710)*, which answers “*what connects Ram and A*?*” without anyone having asked that question in advance.
- **Close on the limitation**, because the question says *qualities*: nothing in this net can say “*no student takes more than six electives*” or “*AI is not a lab course*”. No quantifiers, no negation.

5.N.10 Converting a semantic net into a frame

Compare semantic net and frame. How do you convert a semantic net into a frame? Explain with example 3+5

PIN 1/41 81 Ba 3+5 · the comparison table is \$5.6 of the theory · worked here on the Pee-wee-Reese net of \$5.N.2, because that is the set the textbook itself converts, and both halves are published figures

QUESTION AS PRINTED 81 Ba · Q6 · 3+5

Compare semantic net and frame. How you convert semantic net into frame? Explain with example.

The five steps, applied

- **1. Every node becomes a frame.** The net of §5.N.2 has seven nodes, so the frame system has seven frames: *Person*, *Adult-Male*, *ML-Baseball-Player*, *Fielder*, *Pee-Wee-Reese*, *ML-Baseball-Team*, *Brooklyn-Dodgers*.
- **2. Every outgoing arc becomes a slot.** Arc label $\rightarrow$ slot name, arc target $\rightarrow$ filler. *Adult-Male* height $\rightarrow$ 5-10 becomes a height slot in the *Adult-Male* frame with filler 5-10.
- **3. isa and instance become the class link**, not ordinary slots. *Fielder* gets
isa: *ML-Baseball-Player* and is a subclass frame; *Pee-Wee-Reese* gets instance: *Fielder* and is an instance frame.
- **4. A filler that has its own frame stays a pointer.** *Pee-Wee-Reese* team $\rightarrow$ *Brooklyn-Dodgers* is a link to that frame, not the string. **This is what makes the result a frame system rather than seven separate tables.**
- **5. Push each property to the highest class that has it**, and let everything below inherit. height sits on *Adult-Male*, batting-average on *ML-Baseball-Player*, and *Pee-Wee-Reese* overrides both with his own values.

Then check, and say you checked

- **No information may be lost.** Count the arcs in the net, count the filled slots in the frames. They must agree.
- **What the frames gained:** cardinality (6,000,000,000 persons), defaults (handed: Right), and slots marked with * for values that belong to the class rather than to each member. None of those can be drawn in a net.

5.N.11 Semantic net and frame with FOPL statements

Why are semantic networks and frames important in AI? Provide examples of both with FOPL statements 2+6

PIN 1/41 75 Ba 2+6 · **three formalisms in one answer** — net, frame and FOPL, all saying the same thing · the 2-mark opener is the *why they matter* list in §5.6 of the theory

QUESTION AS PRINTED 75 Ba · Q6 · 2+6

Why semantic network and frames are important in AI? Provide examples of both with FOPL statements example.

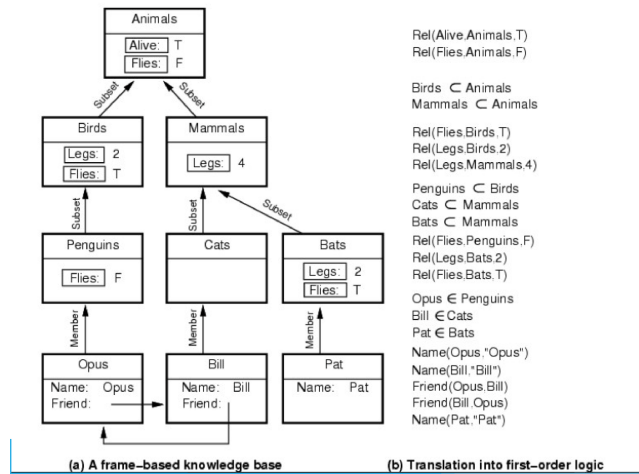
<i>Person</i>	
isa :	<i>Mammal</i>
cardinality :	6,000,000,000
* handed :	Right
<i>Adult-Male</i>	
isa :	<i>Person</i>
card. ty :	2,000,000,000
* height :	5-10
<i>ML-Baseball-Player</i>	
isa :	<i>Adult-Male</i>
cardinality :	624
* height :	6-1
* bats :	equal to handed
* batting-average :	.252
* team :	
* uniform-color :	
<i>Fielder</i>	
isa :	<i>ML-Baseball-Player</i>
cardinality :	376
* batting-average :	.262
<i>Pee-Wee-Reese</i>	
instance :	<i>Fielder</i>
height :	5-10
bats :	Right
batting-average :	.309
team :	<i>Brooklyn-Dodgers</i>
uniform-color :	Blue
<i>ML-Baseball-Team</i>	
isa :	<i>Team</i>
cardinality :	26
* team-size :	24
* manager :	
<i>Brooklyn-Dodgers</i>	
instance :	<i>ML-Baseball-Team</i>
team-size :	24
manager :	Leo-Durocher
players :	{ <i>Pee-Wee-Reese</i> ,...}

Fig. 9.5 A Simplified Frame System

The net of §5.N.2 after conversion. Seven frames, each slot either an inherited class link, a value, or a pointer to another frame. The * marks a value stored on the class. Compare with `c5_semnet_peewee` above: same knowledge, different formalism. Source: PS Sir ch5, after Rich and Knight Fig 9.5.

The reverse direction, if asked

- Frame $\rightarrow$ net is step 1 to 4 run backwards: one node per frame, one arc per filled slot, and isa / instance arcs from the class links.
- **The conversion is lossy in this direction.** Defaults, cardinalities, type restrictions and demons have no arc to become, so they are simply dropped. Say so — it is the cleanest one-line proof that a frame carries more than a net.



(a) the frame-based knowledge base and (b) the first-order sentences that say the same thing. The semantic net is the same picture with the boxes redrawn as nodes and the **Subset** / **Member** labels put on the arcs. Source: PS Sir ch5.

The three-way translation table to reproduce

Net arc	Frame	FOPL
$A \xrightarrow{isa} B$	isa: B in frame A	$\text{Subset}(A, B)$, ie $A \subset B$
$A \xrightarrow{instance} B$	instance: B in frame A	$\text{Member}(A, B)$, ie $A \in B$
$A \xrightarrow{R} v$ on an individual	slot R with filler v	$\text{Rel}(R, A, v)$, ie $R(A, v)$
$A \xrightarrow{R} v$ on a class	slot R on the class frame	$\forall x (x \in A \Rightarrow R(x, v))$

- Worked on the figure's own data: $\text{Subset}(\text{Birds}, \text{Animals})$, $\text{Subset}(\text{Mammals}, \text{Animals})$, $\text{Member}(\text{Opus}, \text{Penguins})$, $\text{Rel}(\text{Legs}, \text{Birds}, 2)$, $\text{Rel}(\text{Legs}, \text{Mammals}, 4)$, $\text{Rel}(\text{Flies}, \text{Penguins}, F)$, $\text{Friend}(\text{Opus}, \text{Bill})$.
- The answer structure that scores:** draw the small net, write the same knowledge as two or three frames beside it, then write one FOPL line per arc underneath. Three formalisms, one set of facts, and the marks are for showing they are interchangeable.
- Close with what FOPL has that neither picture does:** quantifiers, negation and disjunction – and what the pictures have that FOPL does not: inheritance for free, defaults, and a shape a human can check by eye. That is the “why they are important in AI” half of the question.

Coverage. All 12 papers that set a sentence-set conversion are worked above: 72 Ma, 78 Po (§5.N.1); 73 Bh, 79 Ch (§5.N.2); 82 Bh, 81 Ch, 79 Bh (§5.N.3); 82 Ka (§5.N.4); 81 Bh (§5.N.5); 82 Ba (§5.N.6); 81 Ash (§5.N.7); 77 Ch (§5.N.8). §5.N.9 to §5.N.11 work the three remaining drawing questions: the open-ended net 72 Ma, the net-to-frame conversion 81 Ba, and the net/frame/FOPL question 75 Ba. · **Three papers print a defect in their own sentence list**, all reproduced above as printed and all flagged: 78 Po names Sakti Gauchan in a Sandeep Lamichane set, 82 Ka leaves a stray “10.” inside a sentence, and 81 Bh calls the dog Figo once and Fido twice. · **Nothing in this file is machine-checkable:** a semantic net has no arithmetic to recompute, so every arc list above was checked by hand against the OCR archive, sentence by sentence, and the arc counts are printed with each answer so the reader can check them too.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = CT 653 (BCT, III-II, 26 papers) · monospace = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.